

Introducing Google Colab & the Gemini AI Agent

A step-by-step guide covering slides 12–14: opening Colab, creating a new notebook, using the built-in Gemini AI agent, and generating and running Python code.

Activity is designed by Su Nguyen; Detailed guide is drafted by Claude Cowork.

Before you start

You need a Google account. Colab runs in your web browser — nothing to install. The AI features (the Gemini agent) require you to be signed in to Google.

Slide 12	Open Colab and create a new notebook
Slide 13	Meet the Gemini AI agent built into Colab
Slide 14	Generate Python code with AI, then Accept & Run it
Outcome	A working notebook that prints the squares of 1–10

A companion screen recording — [Week2_Colab_Walkthrough_Slides12-14.gif](#) — shows every click in sequence and pairs with the numbered steps below.

Step 1 · Open Google Colab

In your browser, go to the address below and sign in with your Google account if prompted.

<https://colab.research.google.com/>

Colab opens on the “Welcome To Colab” page and shows an Open notebook dialog listing your recent notebooks.

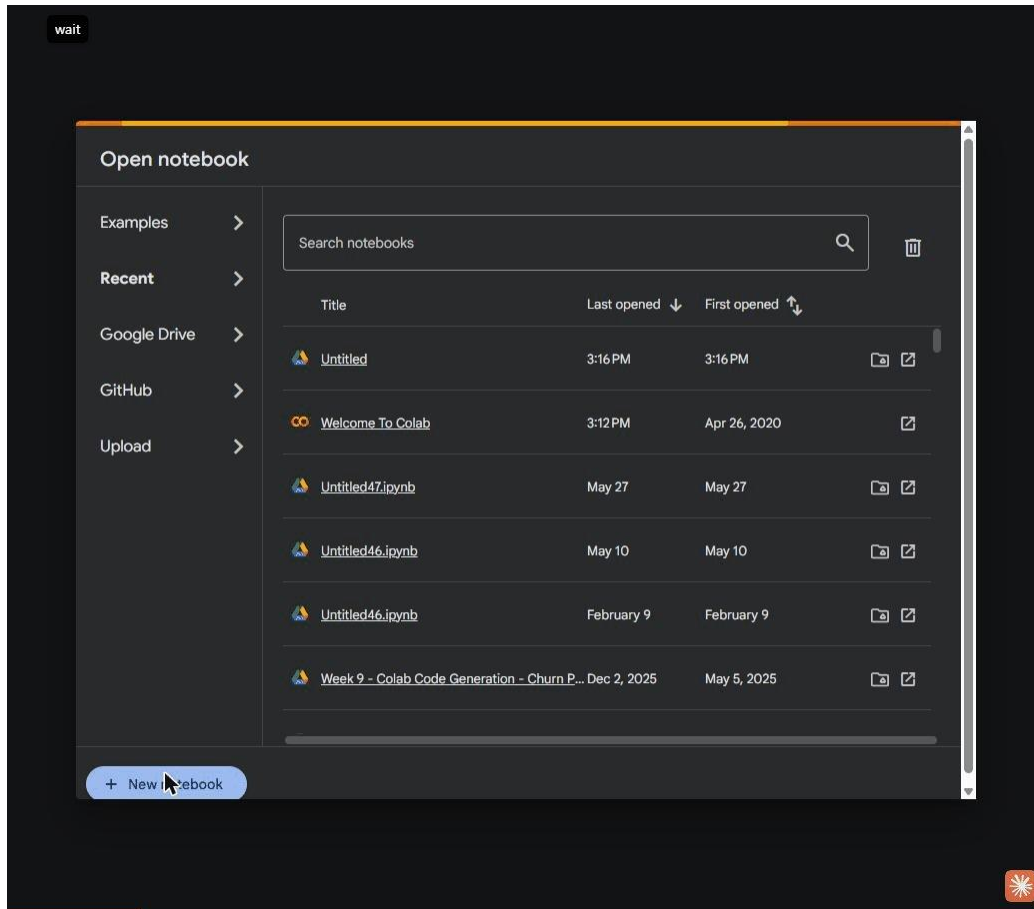


Figure 1 — The Colab landing page with the Open notebook dialog. Click + New notebook (bottom-left).

Step 2 · Create a new notebook (Slide 12)

1. At the bottom-left of the Open notebook dialog, click + New notebook.
2. A fresh notebook opens, automatically named UntitledNN.ipynb (e.g. Untitled48.ipynb). You can rename it later via File → Rename.
3. You will see one empty code cell with the placeholder text: “Start coding or generate with AI.”

Tip: If the Open notebook dialog reappears, press Esc or click Cancel to reach the empty notebook.

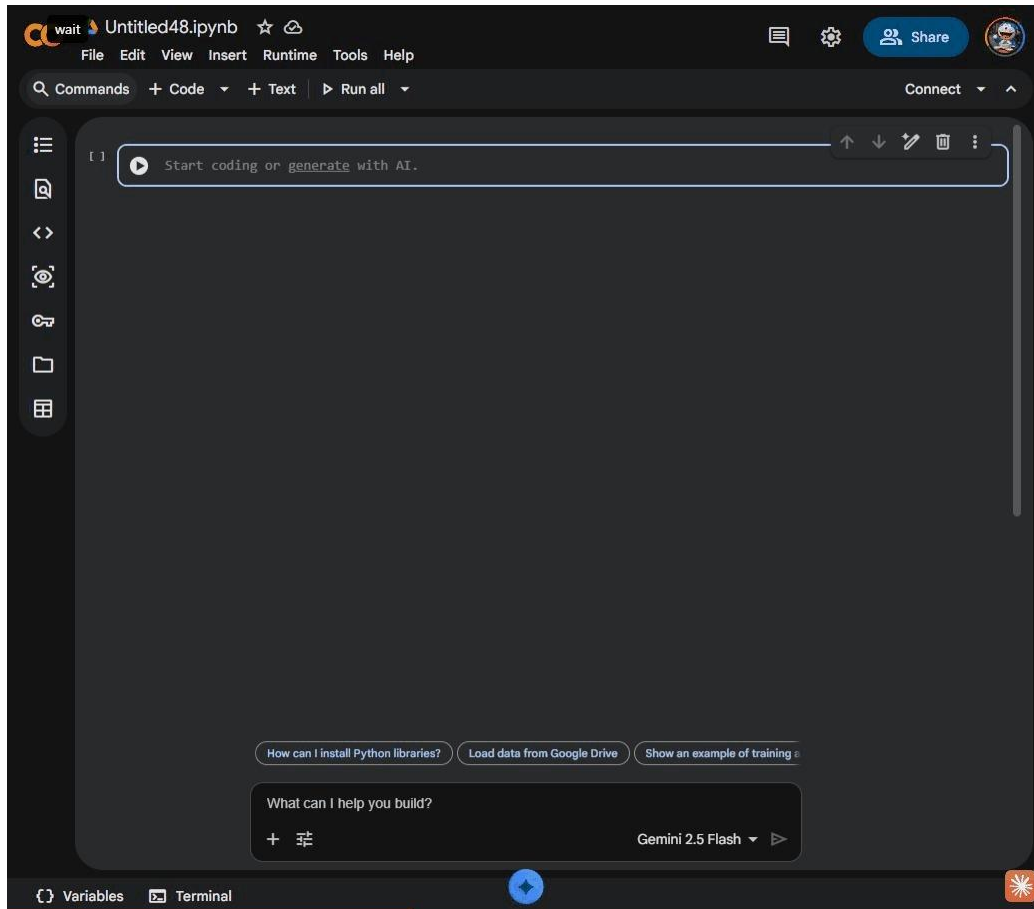


Figure 2 — A new notebook (Untitled48.ipynb): an empty code cell on top and the Gemini chat panel at the bottom.

Step 3 · Meet the Gemini AI agent (Slide 13)

Colab has a built-in AI assistant powered by Gemini. It behaves like ChatGPT or Val — you ask it questions in plain English and it helps you write code.

Where to find it:

- A chat panel sits at the bottom of the notebook with the prompt “What can I help you build?”
- The model in use is shown on the right (e.g. Gemini 2.5 Flash).
- Quick-start suggestion chips appear above it, such as “How can I install Python libraries?” and “Load data from Google Drive”.
- The blue diamond button at the bottom toggles the Gemini panel on and off.

Step 4 · Ask the agent to generate Python code (Slide 14)

Click into the “What can I help you build?” box and type a plain-English request. For this walkthrough:

Generate Python code to print the numbers 1 to 10 and their squares

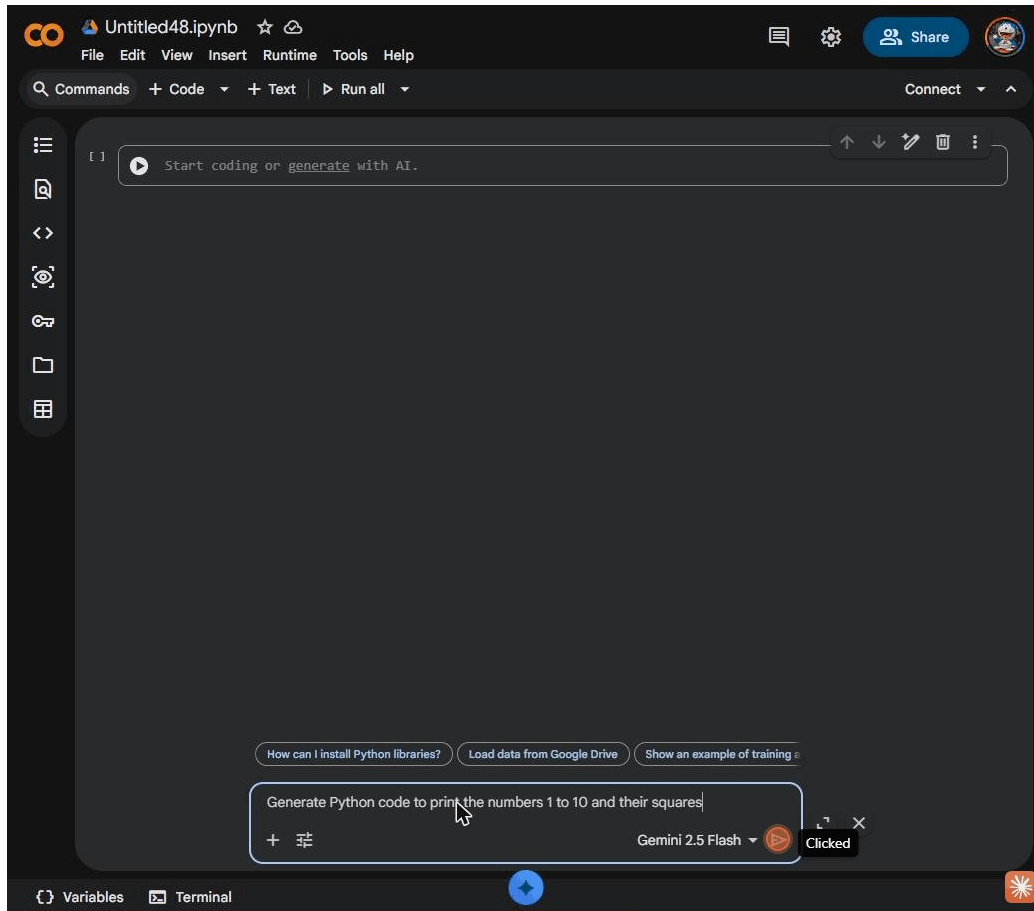


Figure 3 — The request typed into the Gemini panel (model: Gemini 2.5 Flash). Press Enter or click the send arrow.

Press Enter or click the send arrow. Gemini replies with a short explanation and inserts a code cell labelled “Gemini” containing:

```
for i in range(1, 11):  
    print(f"The square of {i} is {i*i}")
```

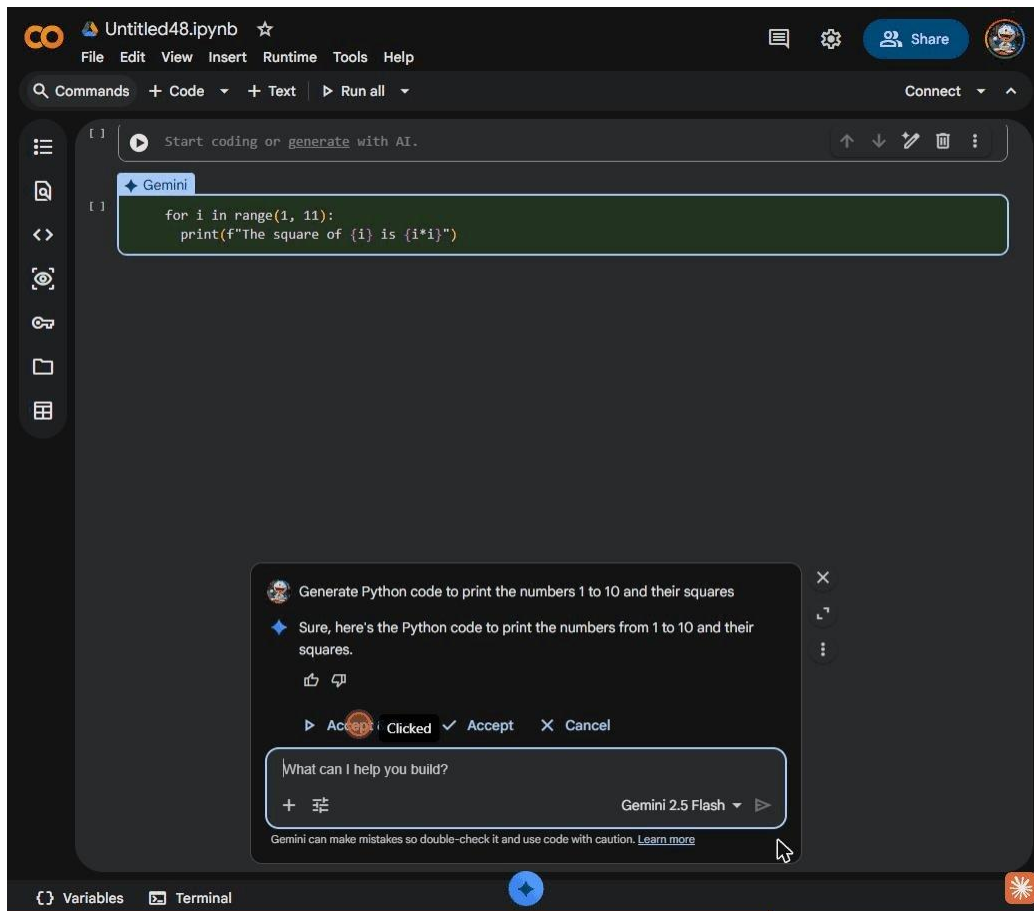


Figure 4 — Gemini inserts a code cell and offers *Accept & Run*, *Accept*, or *Cancel* below its reply.

Step 5 · Accept & Run (Slide 14)

Under the AI response you get three choices:

Accept & Run	Inserts the code AND executes it immediately.
Accept	Inserts the code into the cell without running it.
Cancel	Discards the suggestion.

Click **Accept & Run**. The first time you run anything, Colab connects to a hosted Python 3 runtime — the status at the top-right changes from **Connect** → **Connecting...** → **Connected** (a green tick).

Step 6 · Check the output

The cell runs (note the [1] marker and the 0s timing on the left) and prints:

```
The square of 1 is 1
The square of 2 is 4
The square of 3 is 9
The square of 4 is 16
The square of 5 is 25
```

```
The square of 6 is 36
The square of 7 is 49
The square of 8 is 64
The square of 9 is 81
The square of 10 is 100
```

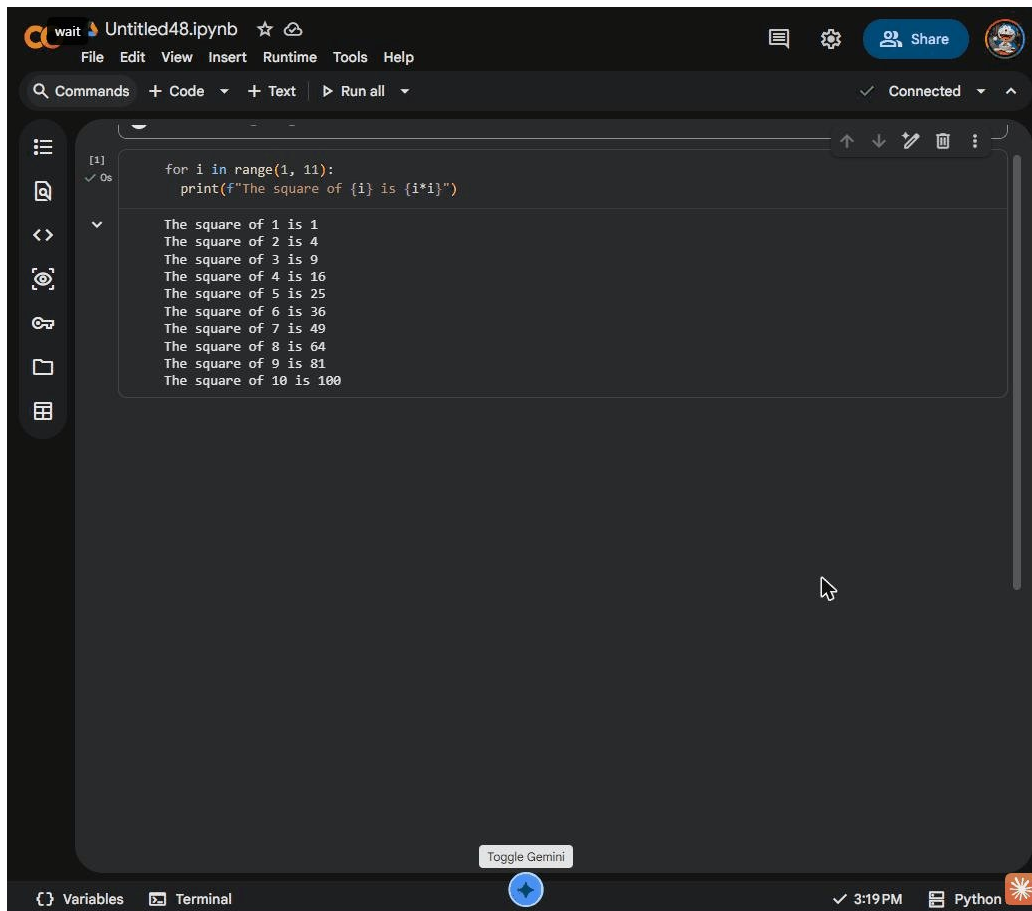


Figure 5 — The cell runs (marker [1], runtime Connected) and prints the squares of 1 to 10.

That is the full loop from slides 12–14: a new notebook, an AI-generated snippet, accepted, run, and producing output — all without writing a line of code by hand.

Good to know

- Always review AI code before trusting it. Colab itself warns: “Gemini can make mistakes so double-check it and use code with caution.”
- Re-run any cell with the round play button on its left, or use Runtime → Run all.
- The runtime is temporary. If it disconnects or you leave it idle, you may need to reconnect and re-run your cells.
- Try your own prompts — e.g. ask it to load a CSV, make a chart, or summarise a dataset. This is the same technique used later to analyse the Melbourne CBD data (slides 15–17).